

ICT1242 Web Development Practicum

Group 05 – Database Design Document

Hospital Appointment, Admission, and Billing System

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Total tables	17 tables
Document	Advanced System – Database Entity Design

1. Overview

This document defines all 17 database tables for the Hospital Appointment, Admission, and Billing System. Each entity section describes the table's purpose, all attributes with data types and key constraints, and its relationships to other tables. The database is designed for MySQL using PHP as the backend scripting language.

The tables are organized into 7 functional groups:

- User & Core – users, patients, doctors
 - Appointment Module – doctor_schedules, appointments, treatment_records
 - Emergency Module – emergency_requests, ambulances
 - Ward & Room Module – rooms, admissions, newborn_records
 - Theatre Module – theatre_operations
 - Pharmacy Module – pharmacy_drugs, prescriptions
 - Billing Module – billing_invoices, billing_items, payments
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2. Entity Definitions

2.1 User & Core Tables

users Stores login credentials for all system users (patients, doctors, reception, pharmacist, admin)			
Attribute	Data Type	Key	Description
user_id	INT AUTO_INCREMENT	PK	Unique user identifier
full_name	VARCHAR(100)		User's full name
email	VARCHAR(100)		Login email (must be unique)
password_hash	VARCHAR(255)		Bcrypt hashed password
role	ENUM('patient','doctor','reception','pharmacist','admin')		Determines access level
phone	VARCHAR(15)		Contact number
created_at	DATETIME		Registration timestamp
is_active	TINYINT(1)		1=active, 0=deactivated account
Relationships: One-to-one with patients or doctors. Referenced by appointments, admissions, billing as created_by.			

patients Extended profile for users with role=patient			
Attribute	Data Type	Key	Description
patient_id	INT AUTO_INCREMENT	PK	Unique patient identifier
user_id	INT	FK	Links to users table
nic	VARCHAR(20)		National ID / passport number
dob	DATE		Date of birth
gender	ENUM('male','female','other')		Patient gender
blood_type	VARCHAR(5)		Blood group (A+, B-, etc.)
address	TEXT		Residential address
emergency_contact	VARCHAR(100)		Name of emergency contact
emergency_phone	VARCHAR(15)		Emergency contact phone
registered_at	DATETIME		When patient was registered
Relationships: Links to users(user_id). Referenced by appointments, admissions, billing_invoices, emergency_requests, prescriptions, newborn_records(mother).			

doctors			
Extended profile for users with role=doctor			
Attribute	Data Type	Key	Description
doctor_id	INT AUTO_INCREMENT	PK	Unique doctor identifier
user_id	INT	FK	Links to users table
specialization	VARCHAR(100)		Medical specialization (e.g. Cardiology)
qualifications	TEXT		Degrees and certifications
license_no	VARCHAR(50)		Medical council license number
consultation_fee	DECIMAL(10,2)		Standard consultation charge
is_available	TINYINT(1)		Whether currently accepting patients
Relationships: Links to users(user_id). Referenced by doctor_schedules, appointments, treatment_records, theatre_operations.			

2.2 Appointment Module Tables

doctor_schedules			
Defines recurring available time slots for each doctor			
Attribute	Data Type	Key	Description
schedule_id	INT AUTO_INCREMENT	PK	Unique schedule entry
doctor_id	INT	FK	Links to doctors table
day_of_week	ENUM('Mon','Tue','Wed','Thu','Fri','Sat','Sun')		Which day this slot applies
start_time	TIME		Slot start time
end_time	TIME		Slot end time
slot_duration_mins	INT		Duration of each appointment slot
is_active	TINYINT(1)		Whether this schedule is currently active
Relationships: Links to doctors(doctor_id). Used by appointment booking to check availability.			

appointments All appointments — both online bookings and OPD walk-ins in one unified table			
Attribute	Data Type	Key	Description
appointment_id	INT AUTO_INCREMENT	PK	Unique appointment identifier
appointment_no	VARCHAR(20) UNIQUE		Auto-generated reference number
patient_id	INT	FK	Links to patients table
doctor_id	INT	FK	Links to doctors table
appointment_date	DATE		Date of appointment
appointment_time	TIME		Time of appointment
source	ENUM('online', 'opd')		Whether booked online or at OPD counter
status	ENUM('pending', 'confirmed', 'completed', 'cancelled')		Current appointment status
notes	TEXT		Patient or booking notes
created_by	INT	FK	user_id of person who created the booking
created_at	DATETIME		Booking creation timestamp
Relationships: Links to patients, doctors, users(created_by). Referenced by treatment_records, billing_invoices.			

treatment_records Doctor notes, diagnosis, and treatment details recorded per appointment			
Attribute	Data Type	Key	Description
record_id	INT AUTO_INCREMENT	PK	Unique treatment record
appointment_id	INT	FK	Links to the appointment
doctor_id	INT	FK	Doctor who recorded this
diagnosis	TEXT		Diagnosis description
notes	TEXT		Doctor's clinical notes
follow_up_date	DATE		Recommended follow-up date
created_at	DATETIME		When the record was created
Relationships: Links to appointments, doctors. Referenced by prescriptions.			

2.3 Emergency Module Tables

emergency_requests Patient ambulance requests submitted via the emergency web page			
Attribute	Data Type	Key	Description
request_id	INT AUTO_INCREMENT	PK	Unique emergency request ID
ticket_no	VARCHAR(20) UNIQUE		Auto-generated emergency ticket reference
patient_id	INT	FK (nullable)	Linked patient profile if registered (can be null for walk-ins)
requester_name	VARCHAR(100)		Name of the person requesting (if not logged in)
phone	VARCHAR(15)		Contact phone for callback
gps_lat	DECIMAL(10,8)		GPS latitude from browser geolocation

gps_lng	DECIMAL(11,8)		GPS longitude from browser geolocation
description	TEXT		Brief description of the emergency
status	ENUM('pending','dispatched','en_route','arrived','closed')		Current request status
ambulance_id	INT	FK (nullable)	Assigned ambulance (set on dispatch)
dispatcher_id	INT	FK (nullable)	user_id of dispatcher who handled request
dispatched_at	DATETIME		When ambulance was dispatched
created_at	DATETIME		When request was submitted

Relationships: Links to patients(optional), ambulances, users(dispatcher). May trigger an admission record on arrival.

ambulances Hospital ambulance fleet and driver details			
Attribute	Data Type	Key	Description
ambulance_id	INT AUTO_INCREMENT	PK	Unique ambulance ID
vehicle_no	VARCHAR(20)		Registration / vehicle number
driver_name	VARCHAR(100)		Driver's full name
driver_phone	VARCHAR(15)		Driver contact number
status	ENUM('available','dispatched','maintenance')		Current availability
last_location	VARCHAR(255)		Last known location or GPS coordinates

Relationships: Referenced by emergency_requests(ambulance_id).

2.4 Ward and Room Module Tables

rooms			
All hospital rooms across all types (semi-private, private, children, ICU)			
Attribute	Data Type	Key	Description
room_id	INT AUTO_INCREMENT	PK	Unique room identifier
room_number	VARCHAR(10)		Room number label (e.g. A-101)
room_type	ENUM('semi_private','private','children','icu')		Category of room
floor	INT		Floor number
bed_count	INT		Total beds in room
occupied_beds	INT		Currently occupied beds
daily_rate	DECIMAL(10,2)		Charge per day for this room type
is_available	TINYINT(1)		Whether room has available beds
Relationships: Referenced by admissions(room_id) and newborn_records(room_id for children's room).			

admissions			
Records of patients admitted to a room including diet and discharge			
Attribute	Data Type	Key	Description
admission_id	INT AUTO_INCREMENT	PK	Unique admission record
patient_id	INT	FK	Admitted patient
room_id	INT	FK	Assigned room
doctor_id	INT	FK	Responsible doctor
admitted_by	INT	FK	user_id of reception staff who processed admission
admission_date	DATETIME		Date and time of admission
discharge_date	DATETIME		Date and time of discharge (null if still admitted)
diet_type	ENUM('standard','doctor_prescribed','none')		Patient meal plan type
diet_notes	TEXT		Specific dietary instructions from doctor
status	ENUM('active','discharged')		Current admission status

notes	TEXT	Admission notes
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Relationships: Links to patients, rooms, doctors, users. Referenced by billing_invoices and theatre_operations(post_ward).

newborn_records Maternity and labour records linking baby profile to mother and operation			
Attribute	Data Type	Key	Description
newborn_id	INT AUTO_INCREMENT	PK	Unique newborn record
patient_id	INT	FK	Patient profile created for the newborn
mother_patient_id	INT	FK	Mother's patient_id
operation_id	INT	FK	Linked theatre operation (labour)
dob	DATETIME		Date and time of birth
weight_kg	DECIMAL(4, 2)		Birth weight in kilograms
birth_notes	TEXT		Medical notes at birth
health_status	ENUM('stable', 'monitoring', 'critical')		Newborn health status indicator
room_id	INT	FK	Children's room assigned to newborn

Relationships: Links to patients(baby), patients(mother), theatre_operations, rooms(children's). Creates a new patient profile for newborn.

2.5 Theatre Module Table

theatre_operations Scheduled surgical and labour operations including theatre, surgeon, and post-op details			
Attribute	Data Type	Key	Description
operation_id	INT AUTO_INCREMENT	PK	Unique operation ID
patient_id	INT	FK	Patient undergoing operation
lead_surgeon_id	INT	FK	doctor_id of the lead surgeon
anaesthetist_id	INT	FK (nullable)	doctor_id of the anaesthesiologist
operation_type	VARCHAR(150)		Type of operation (e.g. Appendectomy, Caesarean)

scheduled_at	DATETIME		Scheduled date and time
duration_mins	INT		Estimated operation duration
theatre_no	INT		Which theatre room is used
pre_op_notes	TEXT		Pre-operation preparation instructions
post_op_notes	TEXT		Post-operation notes from surgeon
status	ENUM('scheduled', 'in_progress', 'completed', 'cancelled')		Operation status
post_ward_room_id	INT	FK (nullable)	Room patient is transferred to after operation

Relationships: Links to patients, doctors(surgeon), doctors(anaesthetist), rooms(post-op ward). Automatically creates billing items for theatre fees. Referenced by newborn_records for maternity operations.

2.6 Pharmacy Module Tables

pharmacy_drugs Master drug catalogue with stock levels and pricing			
Attribute	Data Type	Key	Description
drug_id	INT AUTO_INCREMENT	PK	Unique drug identifier
name	VARCHAR(150)		Drug name (generic or brand)
category	VARCHAR(80)		Drug category (e.g. Antibiotic, Analgesic)
unit	VARCHAR(30)		Unit of measurement (tablet, ml, vial)
unit_price	DECIMAL(10, 2)		Price per unit
stock_qty	INT		Current stock quantity
reorder_level	INT		Minimum quantity before alert is triggered
is_active	TINYINT(1)		Whether this drug is currently available

Relationships: Referenced by prescriptions(drug_id). Stock is decremented on dispensing.

prescriptions Doctor-issued drug prescriptions linked to treatment records and dispensed by pharmacist			
Attribute	Data Type	Key	Description
prescription_id	INT AUTO_INCREMENT	PK	Unique prescription ID
record_id	INT	FK	Links to treatment_records

patient_id	INT	FK	Patient for whom prescription is issued
drug_id	INT	FK	Drug prescribed
dosage	VARCHAR(80)		Dosage instructions (e.g. 500mg)
frequency	VARCHAR(80)		Frequency of administration (e.g. twice daily)
duration_days	INT		Number of days to take the medication
qty_dispensed	INT		Quantity actually dispensed by pharmacist
dispensed_by	INT	FK (nullable)	user_id of pharmacist who dispensed
dispensed_at	DATETIME		Timestamp of dispensing
status	ENUM('pending', 'dispensed', 'cancelled')		Prescription status

Relationships: Links to treatment_records, patients, pharmacy_drugs, users(dispensed_by). Generates a billing_items entry automatically on dispensing.

2.7 Billing Module Tables

billing_invoices Master invoice per patient visit or admission — auto-populated from all modules			
Attribute	Data Type	Key	Description
invoice_id	INT AUTO_INCREMENT	PK	Unique invoice identifier
invoice_no	VARCHAR(20) UNIQUE		Auto-generated invoice number
patient_id	INT	FK	Patient this invoice belongs to
admission_id	INT	FK (nullable)	Linked admission (for inpatient bills)
appointment_id	INT	FK (nullable)	Linked appointment (for outpatient bills)
total_amount	DECIMAL(10,2)		Sum of all billing_items
paid_amount	DECIMAL(10,2)		Total amount paid so far (inc. advances)
balance	DECIMAL(10,2)		Remaining unpaid balance
status	ENUM('pending', 'partial', 'settled')		Payment completion status
created_at	DATETIME		Invoice creation timestamp
created_by	INT	FK	user_id of staff who created invoice

Relationships: Links to patients, admissions(optional), appointments(optional), users. Referenced by billing_items and payments.

billing_items

Individual line items within an invoice — auto-added by each module

Attribute	Data Type	Key	Description
item_id	INT AUTO_INCREMENT	PK	Unique line item ID
invoice_id	INT	FK	Parent invoice
description	VARCHAR(200)		Item description (e.g. Amoxicillin 500mg x 14)
item_type	ENUM('consultation','room','theatre','pharmacy','lab','other')		Category of charge
qty	INT		Quantity
unit_price	DECIMAL(10,2)		Price per unit
total	DECIMAL(10,2)		qty × unit_price

Relationships: Links to billing_invoices(invoice_id). Created automatically by appointment, admission, theatre, and pharmacy modules.

payments

Individual payment transactions — supports both full and advance payments

Attribute	Data Type	Key	Description
payment_id	INT AUTO_INCREMENT	PK	Unique payment record
invoice_id	INT	FK	Invoice being paid
amount	DECIMAL(10,2)		Amount paid in this transaction
payment_type	ENUM('full','advance')		Whether this is a full or partial payment
method	ENUM('cash','card','online')		Mode of payment
received_by	INT	FK	user_id of reception staff who received payment
payment_date	DATETIME		Date and time of payment
receipt_no	VARCHAR(20) UNIQUE		Auto-generated receipt number for this payment

Relationships: Links to billing_invoices, users(received_by). Each payment updates the invoice paid_amount and balance.

3. Key Relationships Summary

The following relationships are central to the system's data integrity:

From Table	Relation	To Table
users	1:1	patients (one user = one patient profile)
users	1:1	doctors (one user = one doctor profile)
patients	1:N	appointments (one patient, many appointments)
doctors	1:N	appointments (one doctor, many appointments)
appointments	1:1	treatment_records (one appointment, one record)
treatment_records	1:N	prescriptions (one visit, many drugs prescribed)
patients	1:N	emergency_requests
ambulances	1:N	emergency_requests (one ambulance assigned to many requests over time)
rooms	1:N	admissions (one room, multiple admissions over time)
patients	1:N	admissions (one patient, multiple admissions)
admissions	1:1	billing_invoices (inpatient bill)
theatre_operations	1:1	newborn_records (maternity only)
pharmacy_drugs	1:N	prescriptions
billing_invoices	1:N	billing_items
billing_invoices	1:N	payments